# Measuring Planck´s and Boltzmann´s constants via photoeffect and transistor diffusion current

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By measuring the photoelectric effect in a photocell, we determined Planck´s constant. We observed the photo current which depends on the counter voltage for five different wavelengths. By plotting the breaking voltage against the frequency we calculated Planck´s contant to $h = (6.46 \pm 0.36) \cdot 10^{-34}$ Js. A transistor was used to determine Boltzmann´s constant. Out of the characteristic line of the transistor at three different temperatures, we calculated Boltzmann´s constant to $k_\mathrm{B} = (1.694 \pm 0.013) \cdot 10^{-23}\,\frac{\mathrm{J}}{\mathrm{K}}$.

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## 1 Introduction

Two of tho most important fundamental physical constants are Planck´s and Boltzmann´s constants. Planck´s constant $h$ is fundamental for quantum mechanics where it describes the relation between energy and frequency of an object. It is also the maximum precision of a measurement in the phase space. The Boltzmann constant $k_\mathrm{B}$ describes the proportionaltity of temperature and energy. When describing big thermodynamical systems one needs a statistical approach due to too high number of degrees of freedom. This approach includes the temperature and with this Boltzmann´s constant.
In the modern SI unit system, all fundamental units are defined by a set of fundamental constants such as $h$ and $k_\mathrm{B}$. This leads to exact values for those constants. The goal of the following two experiments is to confirm the values of both constants.

## 2 Measuring Planck´s constant

### 2.1 Photoelectric effect

By radiating light on a negativly charged metal (photo cathode), the metal can emit free electrons. This effect is called the photoelectric effect and it was interpreted by Einstein in 1905. Einstein said, that electromagnetic waves have a particle character (photons) and carry a discret amount of energy. This can be observed in the photoelectric effect.
Electrons are in a bound state in the atom. If a photon with sufficient energy $E_\mathrm{ph} = h\nu$ hits an electron in the bound state, it can be absorbed, giving its energy to the electron. Now the electron has enough kinetic energy to overcome the work function $W_\mathrm{A}$ of the whole metal plate. Due to that, the electron is a free particle with the remaining kinetic energy

$$E_\mathrm{e} = h\nu - W_\mathrm{A}. \tag{1}$$

By applying an opposing electric field you can maesure the photocurrent, which is the number of electrons overcoming the electric field. The photoelectric effect takes place if the photon energy is greater than the work function of the photo cathode. The kinetic energy of the electrons does not depend on the intensity of the light. It only depends on the frequency of the photons and the work function of the photo cathode´s material.

### 2.2 Experimental setup and examination

The experimental setup, which was used to measure the photoelectric effect, is shown in Fig. 1. A wavelength filter is installed between the Hg-vapor lamp and the photocell in order to irradiate the photocell with light of a certain wavelength. The lense is used to focus the beam on the cathode. At the beginning of the measurement, the counter voltage was set to 3 V. The voltage was then reduced in steps and the associated photocurrent was measured. The voltage was measured with an *Agilent* 34405A voltmeter and a digital multimeter *Keithley* 179 TRMS was used for the current. This process was carried out for five different wavelengths.

We plotted the data of all five wavelengths as shown in figure 2 and 6. The uncertanty of the voltages come from the uncertanty of the used measuring decvice,

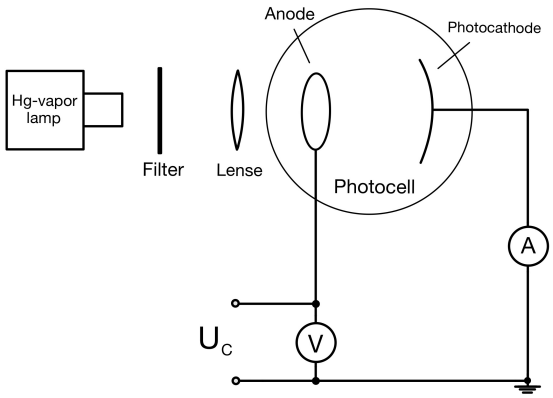


**Fig. 1:** Experimental setup of the photoelectric effect measurement. The photocell is irradiated by the light of the Hg-vapor lamp, which is filtered and focused. The counter voltage is measured by the voltmeter and the photo current is measured by the ammeter.

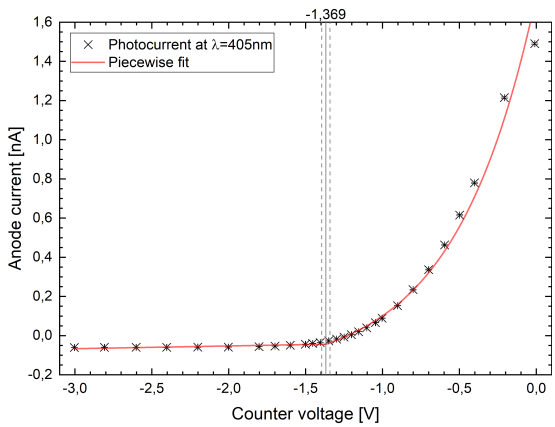


**Fig. 2:** Photo current I in dependance of the applied counter voltage U for light with $\lambda = 405\,\mathrm{nm}$. To determine the braking voltage, a piecewise function is fitted to the data. Below the breaking voltage a linear function and above the breaking voltage an exponential functoin is fitted.

the uncertanty of the current values were approximated by the fluctuations of the displayed current values. These fluctuations appear due to fluctuations in the surrunding magnetic field. For each of the wavelengths we fitted a piecewise function (2) to determine the breaking voltage.This is the counter voltage where the relevant signal is detectable against the background.

$$f(x) = \begin{cases} A + M(x-U) & \text{for } x < U \\ A + B(\mathrm{e}^{C(x-U)-1}) & \text{else} \end{cases} \tag{2}$$

We measure a nonzero background because there is also a opposing photoeffect due to pollutions on the anode in the photocell. This produces an approximatly linear background photocurrent. The determined breaking voltage is the minimum voltage, where the highes energetic electrons out of the photocathode can overcome the applied electric field and reach the anode. At this voltage, we can detect the minimum photocurrent. The breaking voltages shown in tab. 1 are plotted against the fequency $\nu$ of the radiating light. This produces a linear dependency.

| frequency $\nu$ [$10^{14}$Hz] | breaking voltage $U_\mathrm{b}$ [V] |
|---|---|
| $8.191 \pm 0.082$ | $-1.859 \pm 0.024$ |
| $7.402 \pm 0.074$ | $-1.369 \pm 0.027$ |
| $6.876 \pm 0.069$ | $-1.224 \pm 0.018$ |
| $5.491 \pm 0.055$ | $-0.726 \pm 0.017$ |
| $5.187 \pm 0.052$ | $-0.568 \pm 0.011$ |

**Tab. 1:** Breaking voltages for irradiating light of different frequencies. The breaking voltages are determined by a piecewise fit on the $I(U)$ characteristic of the photocell.

Following Einstein´s equation (1), the slope $m$ of the linear fit is

$$m = \frac{h}{e} \tag{3}$$

where $e$ is the elementary charge $e = 1.602176634 \cdot 10^{-19}\,\mathrm{C}$ [2]. The relative error of the frequencies was approximated to 1 %, as there were no information given about the accuracy of used filters. With this method, we receive $h = (6.46 \pm 0.36) \cdot 10^{-34}\,\mathrm{Js}$. The relative error of this measurement is 5.6 %. In it´s margin of error, this result coincides with the exactly defined value $h_\mathrm{exact} = 6.62607015 \cdot 10^{-34}\,\mathrm{Js}$. The relative deviation of the measured from the exatact value is 2.5 %.

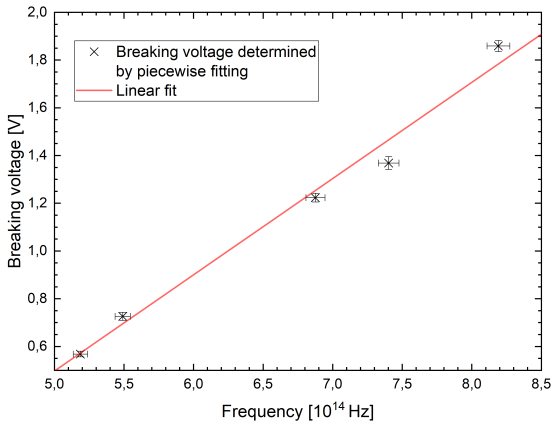


**Fig. 3:** Breaking voltages as the maximum counter voltages where a photocurrent is detectable in dependence of the frequency of the radiating light. The linear fit has a slope $\frac{h}{e}$ according to Einstein´s equation (1).

# 3 Measuring Boltzmann´s constant

## 3.1 Theoretical background

A barrier layer is formed when a p-doped and an n-doped semiconductor meet and an anti-diffusion voltage $U_\mathrm{A}$ develops. Even if no external voltage is applied, two currents are generated in the junction. The intrinsic current $I_0$ is generated by charge carriers, located in the depletion layer, which are thermal excitation into the conduction band. The current follows the potential gradient of $U_\mathrm{A}$. The diffusion current $I_\mathrm{D}$ consists of charge carriers from the p- or n-layer whose energy is large enough to overcome the anti-diffusion voltage. The diffusion current is therefore in the opposite direction to the intrinsic current. If no external voltage is applied, the two currents are equal, so that no external current flows. The number of charge carriers that have a certain energy is described using Boltzmann statistics.

$$N(E)\mathrm{d}E = \mathrm{const} \cdot \mathrm{e}^{-\frac{E}{k_\mathrm{B}T}}\mathrm{d}E \tag{4}$$

Because for the energy of the charge carriers must apply $E > eU_\mathrm{A}$ the diffusion current can be calculated using Boltzmann statistics as:

$$I_\mathrm{D}(U_\mathrm{A}) = \mathrm{const} \cdot ek_\mathrm{B}T\mathrm{e}^{-\frac{eU_\mathrm{A}}{k_\mathrm{B}T}} = |I_0| \tag{5}$$

If an external voltage is now connected in the forward direction, the potential gradient that the charge carriers have to overcome is reduced to $e(U_\mathrm{A} - U)$. This results in the diffusion current:

$$I_\mathrm{D}(U) = \mathrm{e}^{\frac{eU}{k_\mathrm{B}T}} \cdot |I_0| \tag{6}$$

Due to interference currents, a simple p-n-junction is not suitable for measuring the diffusion current. To prevent this, a transistor is used instead of a diode. The two intrinsic currents of the two junctions cancel each other out and since the voltage applied between collector and base is very low, there are hardly any interference currents. Since the diffusion current $I_\mathrm{D}$ flows in the forward direction of the second boundary layer, it passes through it almost undisturbed. From this it follows that the $I_\mathrm{D}$ dependence of the voltage can be measured more precisely.

## 3.2 Experimental setup and examination

To measure the diffusion current $I_\mathrm{D}$ and the base-emitter voltage $U_\mathrm{BE}$ the experiment was constructed acording to Fig. 4. The thermostat *LAUDA* RE 104

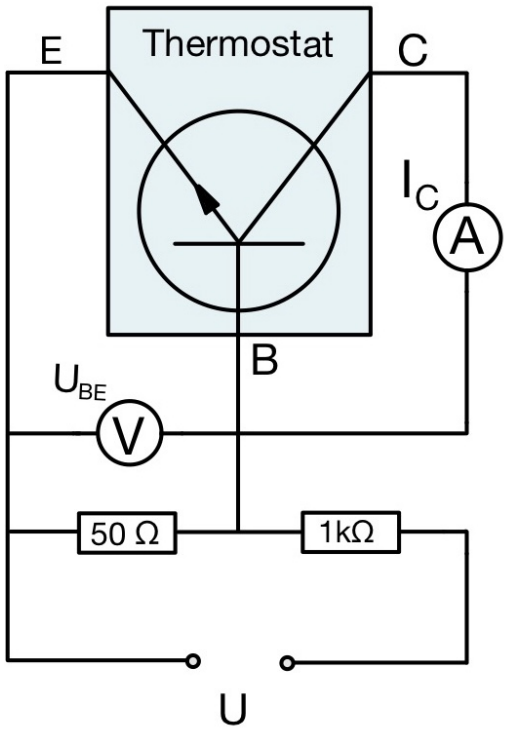


**Fig. 4:** Circuit diagram to measure Boltzmann´s constant. The transistor is kept at a certain temperature by a thermostat. $U_\mathrm{BE}$ and $I_\mathrm{C}$ are then measured using a voltmeter and ammeter. A voltage divider is used to adjust $U_\mathrm{BE}$ more precisely.

was used to bring the transistor to 3 different temperatures. A series of measurements was carried out for each temperature, in which the voltage $U_\mathrm{BE}$ was increased until the diffusion current $I_\mathrm{D}$ reached the value of 10 mA. The voltage was increased in 0.01 V increments and $I_\mathrm{D}$ and $U_\mathrm{BE}$ were measured using the *Agilent* 34405A voltmeters.
The *I-U* dependency was measured for the temperatures $T_1 = 40\,°\mathrm{C}$, $T_2 = 80\,°\mathrm{C}$ and $T_3 = 120\,°\mathrm{C}$. An exponential fit was then applied to the data. An exam-

ple of such a graph is shown in Fig. 5 for $T_1 = 40\,°\mathrm{C}$.

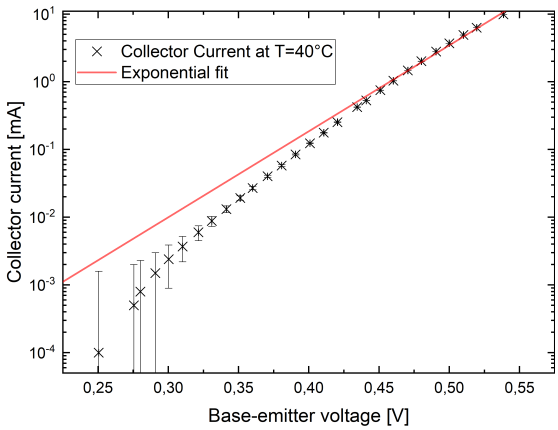


**Fig. 5:** Semi-logarithmic plot of the collector current in dependence of the base- emitter voltage at temperature $T_1 = 40\,°\mathrm{C}$ with an exponential fit function.

The logarithmic plot shows the exponential trend of $I(U)$. Boltzmann´s constant can now be determined from this plot using (6), wich can be rewritten to:

$$\ln(I_\mathrm{D}) = \frac{eU}{k_\mathrm{B}T} + \mathrm{const.} \tag{7}$$

If one determines the slope $m$ of the fit and equates it with $\frac{e}{k_\mathrm{B}T}$, this results in $k_\mathrm{B}$:

$$k_\mathrm{B} = \frac{e}{mT} \tag{8}$$

$k_\mathrm{B}$ was determined for the three measurements at different temperatures.

| temperature $T$ [°C] | Boltzmann´s constant $k_\mathrm{B}$ [$\frac{\mathrm{J}}{\mathrm{K}}$] |
|---|---|
| $40.0 \pm 0.4$ | $(1.750 \pm 0.029) \cdot 10^{-23}$ |
| $80.0 \pm 0.4$ | $(1.713 \pm 0.023) \cdot 10^{-23}$ |
| $120.0 \pm 0.4$ | $(1.653 \pm 0.020) \cdot 10^{-23}$ |
| mean value | $(1.694 \pm 0.013) \cdot 10^{-23}$ |

**Tab. 2:** Values of Boltzmann´s constant measured at different temperatures.

The exact value of Boltzmanns´s constant equates to $k_\mathrm{B} = 1.380649 \cdot 10^{-23}\,\frac{\mathrm{J}}{\mathrm{K}}$ [2]. The value of $k_\mathrm{B}$ determined in this experiment deviates from the exact value by $(22.70 \pm 0.17)\,\%$ and doesn't concur within the limits of its error bar. Based on the residue diagrams in the appendix (Fig. 8) it can be determined that the measured values of $I_\mathrm{D}$ at high voltage deviate from the exponential curve of the $I(U)$ dependency. The reason for this may be the change to the saturation range of the transistor. Therefore the collector voltage no longer has an exponential course. This could explain the deviation to a smaller slope at high voltages which leads to higher values for Boltzmann´s constant.

# 4 Conclusion

The goal of the two previous experiments was to measure Planck´s constant and Boltzmann´s constant. For the determination of Planck´s constant we used the photelectric effect. We measured the photo current in dependence of the counter voltage in a photocell to determine the breaking voltage for five different wavelengths. Our measurements lead to a value of $h = (6.46 \pm 0.36) \cdot 10^{-34}$ Js for Planck´s constant. This value coincides with the exact value of the constant.
The characteristic line of a transistor depends on the temperature and is defined by Boltzmann´s constant. By measuring the characteristic line at three different temperatures, we calculated three values for Boltzmann´s constant. By calculating the mean value, we determined Boltzmann´s constant as $k_\mathrm{B} = (1.694 \pm 0.013) \cdot 10^{-23}\,\frac{\mathrm{J}}{\mathrm{K}}$. This value devates by $(22.70 \pm 0.17)\,\%$ from the exact value. In the data we noticed systematic deviations from the expected characteristic line, which could explain the observed deviations.

# References


[1] Physikalisches Grundpraktikum, Universität Würzburg, Modul C2, Versuch 45, "Plancksches Wirkungsquantum und Boltzmann-Konstante", 2023

[2] P. J. Mohr, D. B. Newell, and B. N. Taylor: "CODATA recommended values of the fundamental physical constants: 2014", Rev. Mod. Phys. 88, 035009 (2016), last visited 06th March, 2023 `https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=920686`

# 5 Appendix

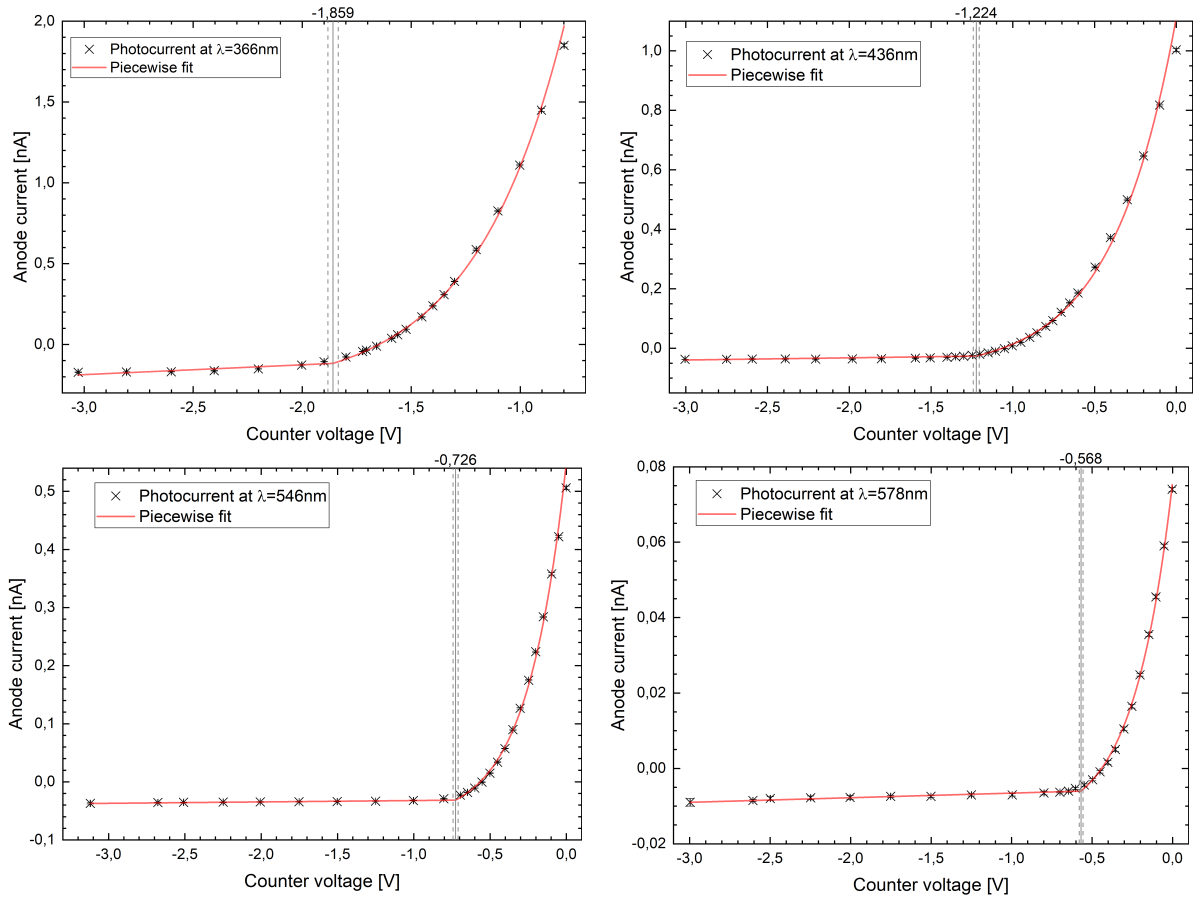


**Fig. 6:** Photo current $I$ in dependance of the applied counter voltage $U$ for light with $\lambda = 366$nm, $\lambda = 436$nm, $\lambda = 546$nm and $\lambda = 578$nm. To determine the braking voltage, a piecewise function is fitted to the data. Below the breaking voltage a linear function and above the breaking voltage an exponential functoin is fitted.

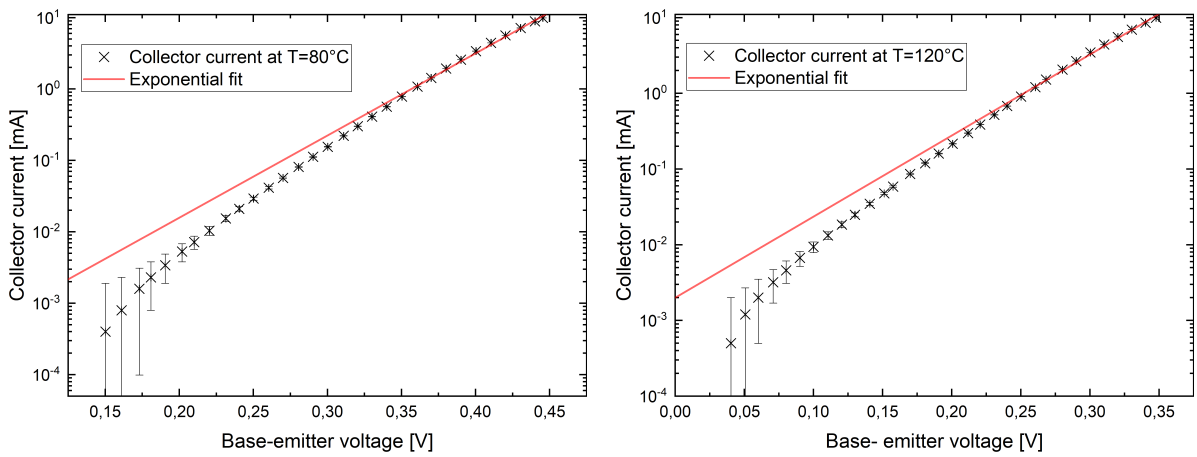


**Fig. 7:** Semi-logarithmic plot of the collector current in dependence of the base- emitter voltage at temperature $T_2 = 80\,°\mathrm{C}$ and $T_2 = 120\,°\mathrm{C}$ with an exponential fit function.

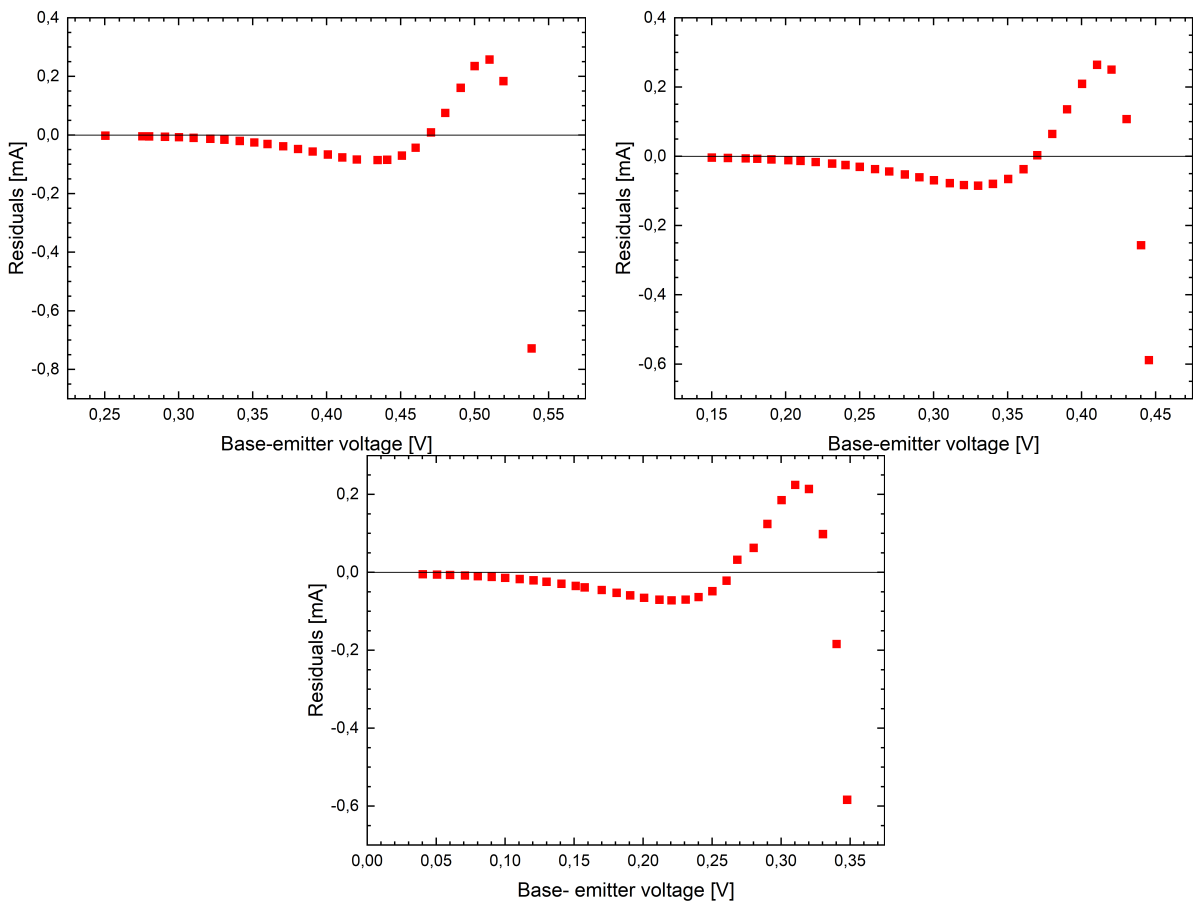


**Fig. 8:** Residual diagrams for an exponential fit applied to the $I_\mathrm{C}(U_\mathrm{BE})$ characteristic curve at $T_1 = 40\,°\mathrm{C}$, $T_2 = 80\,°\mathrm{C}$ and $T_3 = 120\,°\mathrm{C}$.